

Subject Description Form

Subject Code	AMA4325
Subject Title	Derivative Pricing
Credit Value	3
Level	4
Pre-requisite	Introduction to Differential Equations (AMA2008) or Mathematics II (AMA2112) or Further Mathematical Methods for Finance (AMA3723) or Further Mathematical Methods (AMA3724) or equivalent and Probability and Distributions (AMA2691) or Stochastic Processes for Investment (AMA3658) or Principles of Data Science (DSAI1103) [Students are encouraged to enroll AMA3658 prior to AMA4325]
Exclusion	Mathematics for Financial Derivatives (AMA435)
Objectives	This subject is to teach students the basic theory and mathematical techniques for pricing financial options and other derivative securities.
Intended Learning Outcomes	Upon satisfactory completion of the subject, students should be able to: - apply no-arbitrage principle to investigate real-life market discrepancies; - solve elementary stochastic differential equations; - apply the theory of options to consider the pricing of financial derivatives; - synthesize the knowledge and techniques required in solving real-life problems.
Subject Synopsis/ Indicative Syllabus	<i>Introduction to Options and Derivatives</i> Options, forwards, futures, and other derivative securities. (5 hours) Principle of no arbitrage, self-financing strategies, fundamental theorem of asset pricing, risk-neutral valuation, martingale. (8 hours) Properties of stock option prices, put-call parity. (4 hours) <i>Basic Options Theory</i> Option pricing in discrete time: binomial lattice and the Cox-Ross-Rubinstein model. (6 hours) Option pricing in continuous time: geometric Brownian motion and the Black-Scholes formula. (5 hours) Elementary stochastic calculus, Ito's lemma. (8 hours)

	Parameters for the pricing and hedging of options: delta, theta, gamma, vega, rho. (3 hours)					
Teaching/Learning Methodology	The subject will be delivered mainly through lectures and tutorials. The lectures will be conducted to introduce the mathematical concepts for financial derivatives in the syllabus, which are then reinforced by learning activities involving demonstration, tutorial exercise and assignments.					
Assessment Methods in Alignment with Intended Learning Outcomes	Specific assessment methods	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)			
			a	b	c	d
	1. Assignments	20%	✓	✓	✓	✓
	2. Tests	20%	✓	✓	✓	✓
	3. Examination	60%	✓	✓	✓	✓
	Total	100 %				
	Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes: The subject focuses on knowledge, skill and understanding of <u>Derivative Pricing</u> , thus, <u>Exam-based assessment</u> is the most appropriate assessment method, including 20% test and 60% examination. Moreover, 20% worth of assignments are included as a component of continuous assessment so as to keep the students in progress. Continuous Assessment comprises of assignments and tests. A written examination is held at the end of the semester.					
Student Study Effort Expected	Class contact:					
	Lecture			26 Hrs.		
	Tutorial			13 Hrs.		
	Other student study effort:					
	Assignment			40 Hrs.		
	Self-study			30 Hrs.		
	Total student study effort			109 Hrs.		
Reading List and References	<u>Textbook:</u> Hull, J.C. Options, Futures, and Other Derivatives Pearson 2017 10 th edition					
	<u>References:</u> Etheridge, A. A Course in Financial Calculus Cambridge University Press 2002					

	LeRoy, S.F., Werner, J. & Ross, S.A.	Principles of Financial Economics 1 st edition	Cambridge University Press 2000
	Luenberger, D.G	Investment Science 2 nd edition	Oxford University Press 1998
	Panjer, H.H. <i>et al</i>	Financial Economics: With Applications to Investments, Insurance and Pensions	Society of Actuaries 1998
	McDonald, R.L.	Derivative Markets 3 rd edition (Chapters 10-14, 20-24)	Addison Wesley 2009